

IoTRiX 2.0

TRACK C · PROJECT PROPOSAL

Smart Grain Storage and Mushroom Monitoring System

A distributed, low-cost IoT platform for continuous temperature and relative humidity monitoring in Sri Lankan grain storage houses and mushroom cultivation rooms.

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Executive Summary

The Smart Grain Storage and Mushroom Monitoring System proposes an affordable, distributed Internet of Things (IoT) platform for monitoring temperature and relative humidity in Sri Lankan grain storage houses and mushroom cultivation rooms. Multiple sensing locations will provide operators with a clearer view of environmental variation and support timely action when conditions move outside application-specific limits.

Each wireless node uses an ESP32, a DHT11 temperature and relative humidity sensor, and an NTC thermistor. The temperature measurements are processed using Weighted Least Squares (WLS) fusion followed by Kalman filtering. Telemetry is transmitted over Wi-Fi using MQTT to a centralized Node-RED dashboard. The platform will support separate grain storage and mushroom monitoring configurations, with dashboard warnings and local buzzer alerts.

The current proof of concept comprises two operational nodes, each with two temperature sensors, and a working sensing-to-dashboard communication path. A custom PCB and enclosure have been designed; fabrication, full electrical validation, buzzer integration, and application-specific condition assessment remain to be completed. The proposal sets out the methodology, validation activities, implementation milestones, resource requirements, and risks involved in developing the prototype into a deployable system.

1 Introduction and Background

Grain storage and mushroom cultivation require environmental observations that reflect conditions across the monitored space. A reading taken at one location or during an occasional inspection may not represent changes elsewhere in a storage house or growing room. A distributed monitoring system can make these variations visible and provide a continuous record for operational decisions. In stored grain this matters because cereals conduct heat poorly, so local temperature changes and hot spots can remain confined to part of the store. Grain degradation depends mainly on temperature, moisture and oxygen, and the safe moisture level falls as temperature rises; storage fungi require a relative humidity of at least about 65% and grow at roughly 10–40 °C.

For Sri Lankan grain storage houses, the proposed approach uses accessible sensing locations across relevant storage zones. For mushroom cultivation, nodes can be placed in different growing areas or racks. The same hardware platform can serve both applications through different threshold configurations, reducing the need to develop two separate systems.

This proposal focuses on the end-to-end system architecture, sensing and processing method, custom hardware, dashboard, deployment process, and engineering validation relevant to an IoT product submission under Track C. Wireless nodes avoid cabling and give real-time access to readings, so abnormal conditions can be detected and acted on promptly; low-cost wireless sensors have been used in this way in other settings, such as preventive conservation in museums.

2 Problem Statement

Operators need a practical way to identify unsuitable temperature and relative humidity conditions across multiple locations. Periodic manual observations and isolated measurements offer limited visibility of how conditions vary with location and time. This can delay recognition of environmental conditions associated with grain deterioration or unsuitable mushroom cultivation conditions.

The engineering problem is therefore to develop an affordable multi-node platform that acquires dependable measurements, communicates them reliably, and presents application-specific warnings. The system must also distinguish sensor or communication faults from environmental warnings and remain accessible for installation and maintenance.

3 Aim and Objectives

3.1 Project aim

To develop and validate a modular IoT system for continuous, multi-point temperature and relative humidity monitoring in grain storage and mushroom cultivation environments.

3.2 Specific objectives

1. Develop ESP32 sensing nodes with DHT11 and NTC temperature measurements and DHT11 relative humidity acquisition.
2. Calibrate and evaluate the temperature sensors, implement WLS fusion and Kalman filtering, and compare the processed estimate with reference measurements.
3. Provide Wi-Fi and MQTT telemetry to a Node-RED dashboard with unique node identification and connectivity reporting.
4. Implement and validate separate application thresholds, NORMAL, WARNING, and CRITICAL states, and local and dashboard alerts.
5. Fabricate and validate the custom PCB, integrate the enclosure, and assess deployment practicality, power consumption, and communication reliability.

4 Scope of the Project

The current development scope covers the two-node prototype, temperature and relative humidity sensing, edge processing, wireless telemetry, dashboard integration, application configuration, buzzer alerts, and the transition to custom PCB and enclosure hardware. Further nodes may be added after the two-node system is validated.

GPS and the OLED are development aids. The deployed configuration will omit the OLED and retain only temporary GPS connectivity for position initialization where usable. The project does not include direct fungal detection, automated climate control, or current CO₂ measurement. Application thresholds, battery life, field performance, and a maximum supported node count will be established through validation rather than assumed.

5 Proposed Methodology

5.1 System architecture

The system is organized into three operational layers: distributed sensing and edge processing, Wi-Fi and MQTT communication, and centralized monitoring. Each node acquires and processes measurements locally before publishing structured telemetry. The dashboard associates incoming data with a unique node and its configured application. Figure 5.2 in Section 5.5 shows the node-level detail.

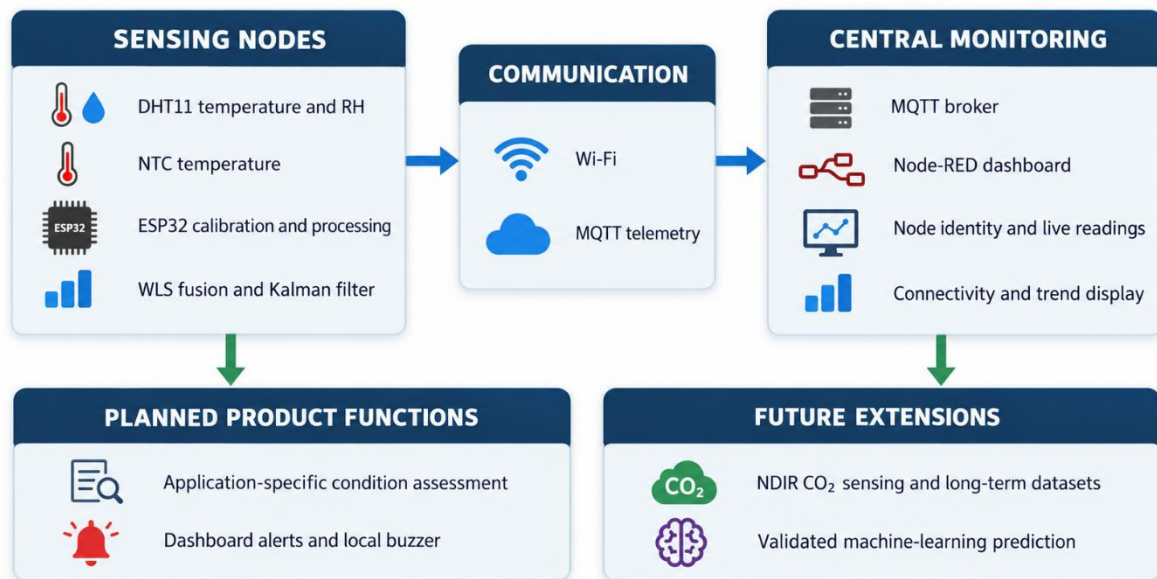


Figure 5.1: Proposed System Architecture and Development Boundaries

5.2 Sensing hardware and product integration

Table 5.1: Hardware Functions

Component	Function and interface
ESP32	Controller for sensor acquisition, processing, Wi-Fi, and telemetry.
DHT11	Digital temperature and relative humidity measurement.
NTC thermistor	Independent temperature measurement through the ESP32 ADC.
Buzzer	GPIO-controlled local alert in the final configuration.
Custom PCB and power circuitry	Integration of the controller, sensors, alert, power, and temporary GPS interface.
Enclosure	Mechanical protection, mounting, ventilation, and access for maintenance.
NEO-6M GPS and OLED	Temporary UART GPS support and an I ² C prototype display; not continuous deployed features.

The PCB and enclosure designs are coordinated around the ESP32 platform. The enclosure must allow sensor exposure, ventilation, connector access, buzzer operation, and maintenance. Published sensor-

node designs show why ventilation matters: air must exchange freely between the housing interior and the room so that the sensor tracks local conditions, using a large aperture and no humidity-absorbing material inside the housing to shorten response time. Where a 3-D printed enclosure is used, material choice should take into account that PLA deformed and degraded under outdoor heat and UV exposure while ASA performed better. Fabrication and physical validation remain pending.

5.3 Sensor processing and conditioning assessment

The node acquires temperature from the DHT11 and NTC thermistor, applies calibration, checks validity, and rejects abnormal measurements. Valid readings are combined using WLS, with relative weights based on experimentally determined measurement uncertainty. The fused result is then processed with a Kalman filter to reduce short-term fluctuation while retaining the underlying trend. DHT11 relative humidity is acquired separately.

Fusion and filtering performance will be evaluated against the individual calibrated sensors. The use of two temperature sensors does not by itself establish higher accuracy; the validation must demonstrate the effect of calibration, weighting, and filtering over representative operating conditions.

In Grain Storage Mode, the proposed condition logic evaluates temperature, relative humidity, and the duration of unfavorable conditions to assign NORMAL, WARNING, or CRITICAL status. In Mushroom Monitoring Mode, the limits depend on the selected mushroom type and cultivation stage. Referenced threshold values and persistence rules must be documented before final validation. For Grain Storage Mode, FAO guidance will be the starting reference: it identifies equilibrium with air at 65–70% relative humidity as the safe condition for long-term storage and relates the permissible storage duration to both temperature and grain moisture content. Environmental warnings indicate a need for inspection or action. They are not confirmation of fungal growth or mycotoxin contamination, which FAO notes can only be established by laboratory examination.

5.4 Communication and dashboard

Nodes publish structured telemetry through Wi-Fi to an MQTT broker, which supplies the Node-RED dashboard. MQTT telemetry to a cloud dashboard has also been used in low-cost open-source environmental sensing hardware. The monitoring interface is intended to show node identity, fused temperature, relative humidity, connectivity, application mode, condition status, live readings, trends, and alarms. Location labels support identification of the corresponding storage zone or growing area.

Reliability work will address Wi-Fi and MQTT reconnection, sensor validity checks, invalid-data rejection, and node connectivity reporting. Temporary buffering during network loss and over-the-air firmware updates are planned improvements. Practical scalability will be assessed using node count, reporting interval, message size, network coverage, broker capacity, and available power. Wi-Fi operates in the 2.4 GHz band, which can struggle over long distances and through thick walls, so coverage will be surveyed at each site before installation.

5.5 Deployment and operation

For grain storage, nodes will be installed at accessible locations representing different storage zones. For mushroom cultivation, locations will represent selected growing zones or racks. Each installation will receive a node ID, a location record, an application mode, and a documented threshold configuration.

Commissioning will proceed through mounting, identification, configuration, network connection, MQTT verification, and dashboard verification before monitoring begins. The prototype GPS supports position initialization where a usable fix is available; reliable indoor GPS positioning is not assumed. The OLED supports development checks and demonstrations and will be removed from the deployed product to reduce hardware cost and power demand.

Figure 5.2 summarizes the complete data flow: the processing performed inside each node, transmission over Wi-Fi and MQTT to the broker and Node-RED, and the application-specific condition assessment that produces the NORMAL, WARNING, or CRITICAL status.

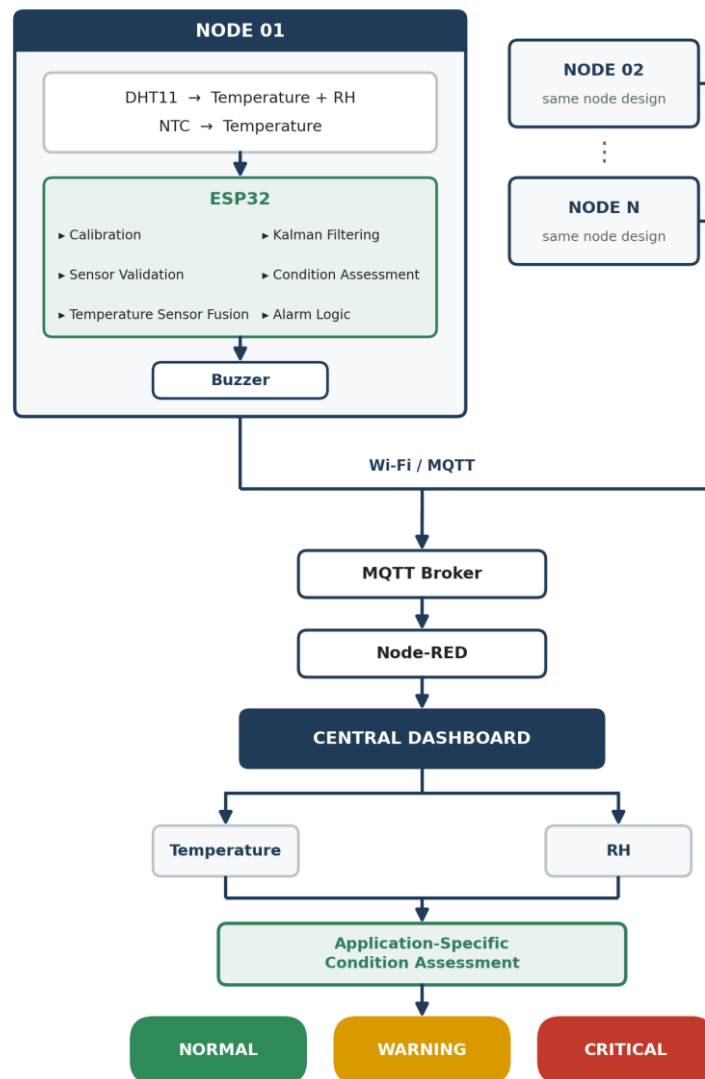


Figure 5.2: Node-Level Processing and System Data Flow

6 Current Progress and Validation Plan

6.1 Current development status

Table 6.1: Current Development Progress

Stage	Present position
Demonstrated or operational	Two sensing nodes; DS18B20 temperature; NTC temperature; WLS fusion; Kalman filtering; Wi-Fi and MQTT; Node-RED dashboard; GPS and OLED development support.
Designed	Custom schematic, routed PCB, 3D PCB representation, enclosure, and final node architecture.
Under development	Buzzer integration, DHT11 integration and calibration, application thresholds and condition logic, PCB fabrication, electrical validation, and deployment validation.
Planned extensions	OTA evaluation, stronger reconnection handling, power characterization, long-duration tests, and broader node deployment.

The prototype demonstrates the end-to-end data path. Quantitative accuracy, latency, reliability, operating duration, and field-performance results remain to be established through the validation plan below.

6.2 Testing and evaluation

Table 6.2: Testing and Evaluation Plan

Validation area	Method and evidence to record
Sensing and processing	Compare DHT11, NTC, fused, and filtered temperature with a reference. Record calibration, error, variability, and response to temperature changes; verify RH readings.
Condition logic and alerts	Apply test values below, near, and above each selected threshold. Verify duration rules, state transitions, dashboard warnings, and buzzer response.
Communication and dashboard	Verify two-node identification, publishing and reception, update delay, disconnections, reconnection, missing data, and displayed values.
PCB and enclosure	Check the fabricated power and signal paths; test sensor interfaces and buzzer; verify assembly, ventilation, connector access, and mounting.
Power and field operation	Measure consumption and operating duration; run longer tests in representative grain storage and mushroom environments.

Test logs, plots, photographs, firmware versions, and identified issues will be retained with the project documentation. Acceptance limits will be defined before the corresponding tests, and completed results will be distinguished from planned activities. Communication reliability will be reported as the proportion of transmitted messages received together with signal strength, as in field evaluations of wireless environmental sensors, and sensing accuracy will be assessed against a reference instrument and an unprotected sensor, as in sensor-shield validation studies.

7 Work Plan and Project Management

7.1 Implementation milestones

Work will proceed through the following milestones. The sequence identifies dependencies; calendar dates will be finalized against the competition schedule, component availability, and PCB fabrication lead time.

Table 7.1: Implementation Milestones

Milestone	Activities	Completion evidence
Finalize requirements	Select and reference application thresholds; define validation limits and alert behavior.	Approved configuration and test plan.
Complete firmware and alerts	Integrate buzzer and application modes; strengthen sensor checks and reconnection handling.	Bench results for sensing, states, and alarms.
Realize product hardware	Fabricate and inspect the PCB; assemble and electrically validate it; fit the enclosure.	Working assembled node and hardware records.
Validate the integrated system	Evaluate sensing, communications, dashboard, power, and longer-duration operation.	Recorded results and resolved fault list.
Conduct representative trials	Install accessible nodes in target environments; evaluate operation and deployment practicality.	Field observations and remaining limitations.
Prepare final submission	Consolidate evidence, documentation, demonstration, and repository.	Traceable technical package and demonstration.

7.2 Project coordination and documentation

Work will be organized across embedded firmware, hardware and mechanical integration, dashboard and communication, and testing and documentation. Named responsibilities and milestone dates will be recorded in the project management documentation.

The GitHub repository will retain the architecture, schematics, PCB and mechanical files, firmware, dashboard flows, communication specification, bill of materials, test data, results, and project documentation. Changes will be committed progressively so that the development history and the tested configurations remain traceable.

The main documentation areas will be docs, hardware, firmware, dashboard, communication, data, results, and project management. Hardware records will include the schematic, PCB, enclosure, and BOM. Demonstration evidence will link the operational nodes to the displayed telemetry and the corresponding firmware revision.

7.3 Final demonstration

The demonstration will show two distinguishable nodes publishing temperature to the dashboard. Completed alert functions will be demonstrated with controlled input conditions. The presentation will distinguish operational features, designed hardware, validation still in progress, and future extensions.

8 Resources and Budget

The platform uses one shared hardware design for both applications, built from commonly available embedded components. All costs are in Sri Lankan rupees (Rs). Section 8.1 records the prototype bill of materials, while Section 8.2 updates the final-product budget using the supplied final PCB BOM. Costs not included in that BOM are clearly identified as TBC.

8.1 Prototype bill of materials

Table 8.1: Semi-final Prototype Bill of Materials

Item	Component	Specification	Qty	Unit cost (Rs)	Total (Rs)
1	ESP32 Dev module	ESP32 Wi-Fi/Bluetooth development board	1	1500.00	1500.00
2	DS18B20 Temperature Sensor	Digital 1-Wire temperature sensor	1	160.00	160.00
3	NTC Thermistor	MF5A-3, 10 kΩ, B3950	1	60.00	60.00
4	GPS Module	NEO-6M GPS module	1	2490.00	2490.00
5	OLED Display	SSD1306, 128 × 64, I2C, 0.96 in	1	940.00	940.00
6	N-Channel MOSFET	IRLZ44N	2	95.00	190.00
7	Push Button	Momentary push button	1	45.00	45.00
8	Fixed Resistor	10 kΩ, for NTC voltage divider	1	5.00	5.00
9	Pull-up Resistor	4.7 kΩ, for DS18B20 DATA line	1	5.00	5.00
10	MOSFET Gate Resistor	220 Ω	2	3.00	6.00
11	MOSFET Gate Pull-down Resistor	100 kΩ	2	6.00	12.00
12	1S Battery Holder	Single-cell battery holder	1	40.00	40.00
13	Rechargeable Battery	1800 mAh 3.7 Li-ion Cell	1	400.00	400.00
14	Soldering Iron Lead	For soldering		Free	-
15	Single core Cu wire	For connections	2m	45.00	90.00
16	10x24 Dot board	Mounting components	1	200.00	200.00
				TOTAL COST	6143.00

The prototype total of Rs 6,143.00 includes Rs 3,720.00 of items that are not part of the final unit: the NEO-6M GPS module and SSD1306 OLED display, which are development aids, and the wire and dot board used for hand wiring.

8.2 Budget for one final-product unit

Table 8.2: Budget for One Final-Product Unit

Component	Specification	Qty	Unit cost (Rs)	Total (Rs)
Capacitor	10 μF, through-hole	2	5.00	10.00
Capacitor	100 nF, through-hole	1	2.00	2.00
4-pin connector	HC-EH2.54-4A-TW	1	40.00	40.00
JST connector	JST-PH-2-SMT-RA	1	20.00	20.00
5-pin header	2.54 mm, 1×5	2	6.00	12.00
N-Channel MOSFET	IRLZ44PBF / TO-220	1	95.00	95.00
10 kΩ resistors	Through-hole	7	1.50	10.50
100 Ω resistor	Through-hole	1	1.50	1.50
Push buttons	4.5×4.5×3.8 mm tactile switch	2	18.00	36.00
ESP-12F module	2.4 GHz Wi-Fi module	1	660.00	660.00
2-pin jumper header	2.54 mm	1	5.00	5.00
Voltage regulator	AMS1117-3.3	1	20.00	20.00
NTC thermistor	10 kΩ	1	60.00	60.00
Buzzer	HX-2212 local alert buzzer	1	55.00	55.00
DHT11 sensor	Temperature and relative humidity sensor	1	230.00	230.00
Rechargeable battery / power source	3500 mAh Li-ion	2	690	1380.00
Custom PCB fabrication	Board fabrication cost	1	TBC	TBC
3-D printed enclosure	Final mechanical enclosure	1	TBC	TBC

Component	Specification	Qty	Unit cost (Rs)	Total (Rs)
CONFIRMED PCB ELECTRONICS SUBTOTAL				2,625.00
TOTAL PER UNIT				Rs 2,625.00 +PCB fabrication + enclosure

Based on the supplied final PCB bill of materials, the confirmed electronic component cost for one unit is Rs 1,245.00. This figure includes the ESP-12F module, DHT11 sensor, NTC thermistor, buzzer, voltage regulator, MOSFET, switches, connectors, capacitors, resistors, and headers used on the final board. The battery or other power source, PCB fabrication, and 3-D printed enclosure are not priced in the supplied BOM and are therefore shown as TBC until their final quotations are available.

The final per-node deployment cost will therefore be the confirmed Rs 1,245.00 electronics BOM plus the selected battery/power source, PCB fabrication, and enclosure costs. A deployment with N nodes will scale approximately with the number of nodes, while shared infrastructure such as Wi-Fi access, the MQTT broker, and the Node-RED host is counted separately where it is not already available.

9 Risks and Mitigation

Table 9.1: Risks and Proposed Mitigation

Risk or limitation	Proposed mitigation
Unvalidated application thresholds	Select referenced limits for each application and verify state and duration rules before field use.
Sensor bias or invalid readings	Calibrate, compare with a reference, validate fusion, and flag rejected or failed measurements.
Wi-Fi or broker interruption	Test reconnection and missing-node reporting; evaluate temporary buffering.
Fabrication or assembly issues	Review the design, inspect fabricated boards, and perform electrical checks before enclosure integration.
Uncharacterized power demand	Measure operating consumption and establish achievable operating duration.
Limited field evidence and node count	Expand beyond the two-node demonstration only after longer, representative tests.

10 Expected Outcomes and Significance

The immediate outcome will be a validated two-node monitoring system with documented temperature processing, relative humidity acquisition, MQTT telemetry, and a centralized dashboard. The product development work is expected to deliver fabricated and tested PCB hardware, integrated enclosures, configurable condition assessment, and local and dashboard alerts.

The proposed value lies in combining distributed measurements, two temperature sensing methods, local processing, and a shared monitoring interface within a modular platform. Separate application

configurations will allow the same core hardware to serve grain storage houses and mushroom growing rooms. Accessible mounting and replaceable components will support practical maintenance.

The intended benefit is improved visibility of environmental conditions and earlier recognition of unsuitable conditions. Reductions in spoilage, improvements in yield, and economic savings will require field evidence and are not claimed as established outcomes. The validation records will determine which performance claims can be made for the final system.

11 Future Development

11.1 Carbon dioxide monitoring

A later version may add a dedicated non-dispersive infrared (NDIR) CO₂ sensor to measure concentration in parts per million. This would extend the environmental data available for grain storage and mushroom cultivation assessments. Sensor selection, calibration, placement, cost, and application interpretation will require separate evaluation. CO₂ monitoring is outside the current prototype scope.

11.2 Historical data and predictive analysis

Long-term data storage will support trend analysis and the development of application-specific datasets. Candidate records include temperature, relative humidity, timestamps, node identity, location, application mode, condition status, and verified observations of grain quality or mushroom growth. Future CO₂ measurements may be added once sensing is implemented.

Machine-learning prediction will be considered only after sufficient representative and correctly labelled data are available. Until then, transparent threshold-based logic provides the basis for condition assessment. Further development may also extend node deployment, local buffering, and validated OTA maintenance.

12 Conclusion

The proposed system provides a practical development path from an operational two-node IoT demonstration to an integrated environmental monitoring product. The next stage will focus on application-specific assessment, alerts, fabricated hardware, reliable communication, and representative validation. Completing these activities will provide the evidence needed to assess deployment readiness and the feasibility of a reusable monitoring platform for Sri Lankan grain storage and mushroom cultivation.

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